

Sickle Cell Diseases and the Kidney

Claire C. Sharpe, Fatiu A. Arogundade

SICKLE CELL DISEASE

Epidemiology

Sickle cell disease (SCD) is an autosomal recessive inherited disorder that predominantly affects persons of African, Mediterranean, Indian, and Middle Eastern descent but was first recognized in West Africa. The high prevalence of SCD in this region probably represents a survival benefit, as the presence of the sickle cell gene protects against malaria. SCD is now a worldwide health problem because the carrier state has spread throughout Africa, around the Mediterranean, and to the Middle East and India, as well as to the Caribbean, North America, and northern Europe. The prevalence of the sickle cell gene is about 8% in African Americans and about 25% in adult Nigerians. Of babies born with SCD, 25% worldwide are born outside of sub-Saharan Africa.¹⁻³

Genetics

The sickle hemoglobin (Hb) mutation (hemoglobin S, or HbS) results in the replacement of the normal glutamine by valine in the sixth position of the β -globin subunit, thereby changing the configuration of the Hb tetramer molecule in the homozygous person from $\alpha_2\beta_2$ to $\alpha_2\beta^S_2$. Sickle cell disease occurs in those homozygous for HbS (commonly referred to as *sickle cell anemia*) or in heterozygotes when HbS coexists with another abnormal or missing β -chain (e.g., HbC [$\alpha_2\beta^S\beta^C$] or HbS β thalassemia [$\alpha_2\beta^S\beta^0$]). Sickle cell trait occurs in those heterozygous for HbS when the other Hb molecule is normal (HbAS [$\alpha_2\beta^A\beta^S$]).

Several HbS haplotypes—mutations of the HbS molecule—have been identified that have probably arisen independently of each other. There are four major types in Africa—the Benin, Senegal, Cameroon, and Bantu (or Central African Republic) haplotypes—and one Asian haplotype. Variations in these haplotypes determine disease severity; for example, the Senegalese haplotype is associated with a higher fetal Hb (HbF) concentration (a key driver of disease severity) and has a better prognosis than others. In a sample population of Nigerians, Benin haplotype was detected in 92%. Sex also influences disease severity; female HbSS patients with Benin haplotype have a higher HbF level than male patients.^{4,5}

Pathophysiology

The driving pathophysiologic factor is HbS polymerization. During cellular or tissue hypoxia, dehydration, or oxidative stress, the mutated β -globin chains of the HbS molecule tend to crystallize to a polymer nucleus. Polymerization disrupts the architecture and changes the shape of the red blood cell (RBC) to a characteristic crescent or sickle, which increases its rigidity (Fig. 51.1). Polymerization is dynamic and depends on local oxygen tension. It is also promoted by acidosis (which decreases the affinity of HbS for oxygen) and hyperosmolality (which increases RBC Hb concentration). As a consequence of repeated

polymerization cycles, sickle RBCs exhibit abnormally high adhesion to activated endothelium, owing to acquired membrane changes and retained adhesion receptors on reticulocytes, especially the stress reticulocytes. This increases microvascular transit time, thereby stimulating further sickling. This results in premature destruction (hemolysis) of the RBCs and frequent, widespread vaso-occlusive episodes with subsequent acute and chronic organ damage.

The tendency of HbS to polymerize is influenced by the presence or absence of HbF, which contains γ -Hb chains in place of β -Hb ($\alpha_2\gamma_2$). In RBCs that contain HbF (known as F cells), the presence of HbF dilutes the concentration of HbS, thus increasing the threshold at which polymerization occurs. In addition, both HbF and its mixed hybrid tetramer ($\alpha_2\beta^S\gamma$) cannot enter the deoxy sickle Hb polymer phase, further limiting the degree of polymerization that can occur within a cell.⁶ Therefore, the main determinant of disease severity is the rate and extent of HbS polymerization, which drives the two major pathophysiologic processes: vaso-occlusion with ischemia-reperfusion injury and hemolytic anemia (Fig. 51.2).⁷

Vaso-occlusion occurs in all patients, often triggered by inflammation, and is typically found in clinical states of infection, hypoxia, hypovolemia, hypothermia, acidosis, and hyperosmolality. Vaso-occlusion is probably caused by dynamic endothelial-leukocyte-RBC adhesive interactions in the postcapillary venules and precapillary obstruction by rigid, deformed RBCs. Episodic microvascular occlusion and ischemia can be interrupted by restoration of blood flow and reperfusion, which further promote inflammatory stress and tissue injury. In addition, hemolysis contributes to the development of progressive vasculopathy, characterized by endothelial dysfunction and proliferative changes in the intima and smooth muscle of blood vessels. An important role is ascribed to free Hb, which inactivates nitric oxide and generates reactive oxygen species. These separate pathologic processes have been used to explain the distinct subphenotypes of clinical complications of SCD whereby some patients present with frequent vaso-occlusive crises (VOCs), acute chest syndrome (ACS), and osteonecrosis, whereas others have more severe anemia, recurrent leg ulceration, pulmonary hypertension, priapism, and chronic kidney disease (CKD).⁸

Natural History and Clinical Manifestations

Sickle cell disease is a highly variable condition, though life expectancy is reduced in all subjects, especially in those with symptomatic disease. Median survival in some parts of sub-Saharan Africa may be as low as 5 years, whereas it is 45 to 55 years in the United States and Jamaica.^{5,9} The clinical manifestations of SCD are individualized and age dependent (Fig. 51.3).¹⁰ Newborn babies are asymptomatic for the first couple of months of life because fetal Hb predominates at this age. However, as γ -chains are rapidly replaced through β -chain synthesis, symptoms (often potentially life-threatening) begin

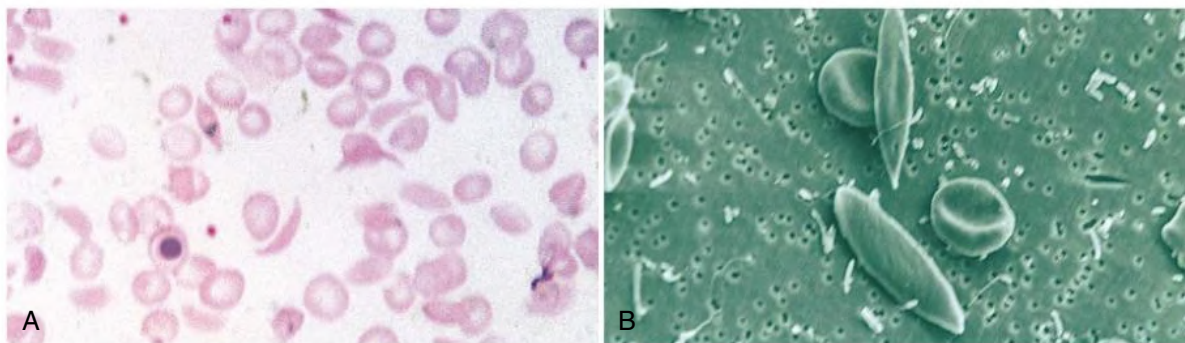


Fig. 51.1 Sick Cell Cells. (A) Characteristic sickle cell erythrocytes in peripheral blood film of a patient with homozygous sickle cell anemia. (B) Electron micrograph showing two normal and two sickle-shaped erythrocytes. (Courtesy Professor Sally C. Davies.)

Pathophysiology of Sick Cell Disease

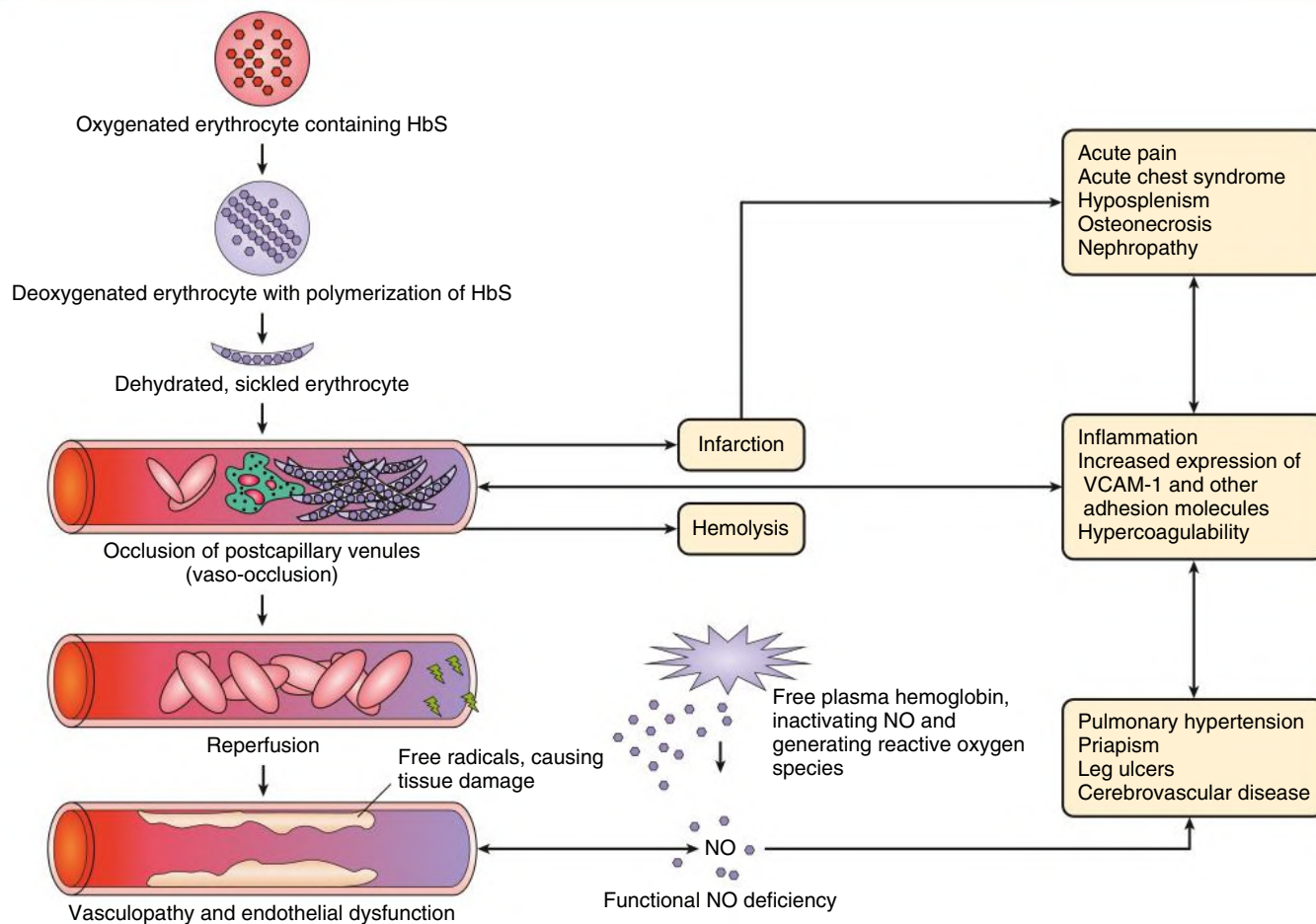


Fig. 51.2 Pathophysiology of sickle cell disease. HbS, Hemoglobin S (sickle Hb); NO, nitric oxide; VCAM-1, vascular cell adhesion molecule 1. (Modified from Rees DC, Williams TN, Gladwin MT. Sickle-cell disease. *Lancet*. 2010;376:2018–2031.)

to occur, including dactylitis, ACS, overwhelming sepsis, and acute splenic sequestration. Stroke is also a feature of early childhood, with an incidence of 8% by age 14 years, rising to 11% by age 20 years. However, this has been largely mitigated in wealthy countries by screening at-risk children with transcranial Doppler (looking for increased blood flow velocity, which is suggestive of arterial stenosis)

and the use of chronic transfusion for both primary and secondary prevention.¹¹ Human parvovirus B19 infection can lead to a severe and sudden fall in Hb in children and adolescents with SCD because the virus destroys RBC precursors for 8 to 10 days, which results in severe anemia. After the age of 5 years, when active bone marrow is withdrawn from the small bones, dactylitis is replaced by the

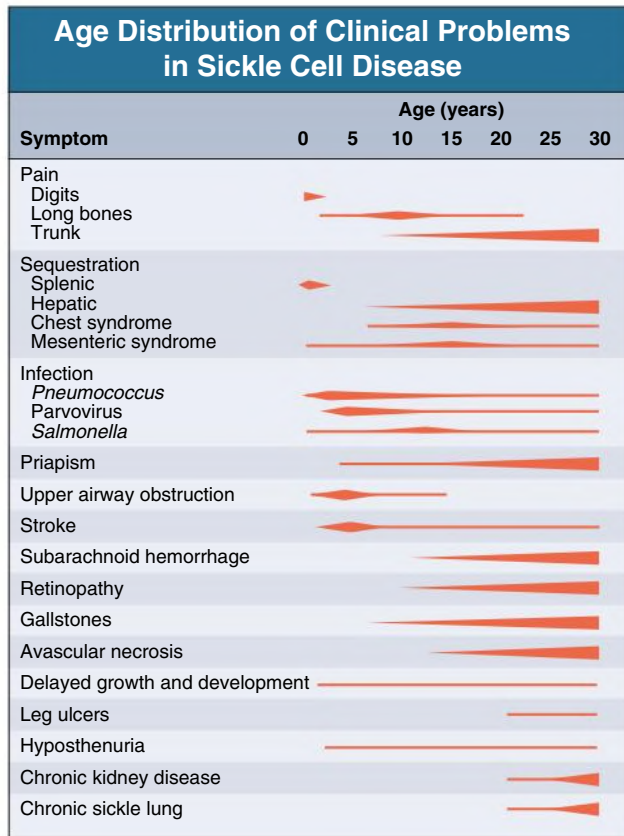


Fig. 51.3 Age distribution of clinical problems in sickle cell disease. (Modified from Davies SC, Oni L. Management of patients with sickle cell disease. *BMJ*. 1997;315:656–660.)

classic painful vaso-occlusive crises, which increases in frequency with increasing age. In addition, adolescence is associated with nocturnal enuresis, avascular necrosis of the hip, leg ulcerations, delayed puberty, and priapism. Over the age of 25 to 30 years, the frequency of VOC tends to reduce and is replaced with signs and symptoms of chronic organ damage, including heart failure, pulmonary hypertension, sickle hepatopathy, and sickle cell nephropathy.⁵ In younger patients, the primary cause of death is usually infection, whereas in older patients the primary cause of death is commonly irreversible organ damage (Fig. 51.4).¹²

SICKLE CELL NEPHROPATHY

Sickle cell nephropathy (SCN) develops slowly, starting in the very young with glomerular hyperfiltration and leading to albuminuria in late childhood or early adulthood. Although many patients do not progress further, a number develop progressive CKD. These patients are at increased risk for acute kidney injury (AKI) complicating VOC or other interim illnesses, events that often precipitate a further decline in glomerular filtration rate (GFR). The prevalence of albuminuria is approximately 60% in those over 45 years, but only 4% to 12% of patients with SCD develop end-stage kidney disease (ESKD), though this figure may increase as the patient cohort ages.^{12,13} In keeping with this, in 2008 advanced CKD was reported in 24% of patients with SCD who had survived to 60 years of age or more and was the cause of death in 45%.¹⁴

In general, SCN is more prevalent and more severe in those with HbSS-thalassemia and HbS β^0 -thalassemia than in those with milder forms of SCD such as HbSC.^{9,15}

Genetic Modifiers of Risk for Developing Sickle Cell Nephropathy

Other genetic modifiers are also associated with disease severity and risk for developing SCN, often through their influence on HbF levels. A polymorphism in the *BCL11A* gene (a fetal Hb silencing factor) leads to higher HbF levels, reduced hemolysis, and amelioration of SCD-related complications. Coinheritance of α -thalassemia and SCD is found in approximately one-third of patients and is associated with reduced hemolysis and protection from albuminuria. Over the past decade, genetic risk factors for CKD have been identified in the African American population, the most important of which are haplotypes of the *APOL1* gene. These are associated with increased risk for hemolysis and, when combined with genotypes for α -thalassemia and *BCL11A*, can be used to stratify risk for progressive CKD in SCD patients.¹⁶ Using estimated GFR as a marker of kidney function, the transforming growth factor- β /bone morphogenic protein (TGF- β /BMP) pathway¹⁷ has been implicated as a cause of progressive kidney function loss in SCN.

Pathophysiology of Sickle Cell Nephropathy

The pathogenesis of SCN is directly related to the blood supply of the kidney and its circulation. In health the kidneys receive approximately 25% of the cardiac output to achieve effective filtration of the plasma at a rate of approximately 100 mL/min/1.73 m², despite representing less than 1% of total body weight. As a consequence, the cortex of the kidney is at risk for receiving excess oxygen, which is mitigated by shunting of oxygen from the afferent arteriole to the efferent arteriole. The vessels (vasa recta) that supply the medulla of the kidney branch off early from the efferent arteriole, taking only a fraction of the total renal blood flow (RBF) with them. Much of the blood that enters the kidney cortex is therefore delivered back to the venous circulation without entering the medulla at all. The maintenance of the relatively sluggish but intricate medullary blood flow is critical to maintaining the corticomedullary interstitial solute gradient generated by the countercurrent multiplier system of the loop of Henle, which drives water and solute reabsorption and allows for effective urinary concentration.¹⁸ The resulting hypoxia (partial pressure of oxygen 10–35 mm Hg), acidosis, and hyperosmolality of the inner medulla make it an ideal environment for the polymerization of deoxygenated HbS and subsequent sickling of erythrocytes. Over time, repeated cycles of sickling and sludging cause microinfarcts and ischemic injury leading to the chronic microvascular disease that is apparent in established SCN (Fig. 51.5).¹⁹ Local activation of hypoxia inducible factor 1 α (HIF1 α) upregulates the expression of endothelin-1 which, in the presence of reduced nitric oxide, leads to an increase in reactive oxygen species and vasoconstriction, thus feeding into a cycle of chronic medullary hypoxia (Fig. 51.6).²⁰

Despite reduced circulation in the kidney medulla, total RBF and GFR are paradoxically increased. The subsequent hyperfiltration eventually results in proteinuria and glomerulosclerosis, which together with tubulointerstitial fibrosis marks the onset of progressive CKD.²¹ Although partially due to the increased cardiac output seen in patients with SCD, this increase in RBF cannot be completely explained by anemia or increased plasma volume, as it is not reversed by correction of the anemia with transfusion. One explanation involves the hemoxygenase–carbon monoxide system. Heme oxygenase-1 (HO-1) is upregulated in injured kidneys (and other tissues) in response to ongoing hemolysis in SCD, and is responsible for the conversion of heme to biliverdin with the subsequent release of carbon monoxide. Both biliverdin and carbon monoxide at these levels are potent antioxidants, and the carbon monoxide acts locally as a vasorelaxant, thus increasing both RBF and GFR.²² This provides a possible link between the degree of hemolysis experienced by a patient and the likelihood of developing proteinuria.

Mortality in Sickle Cell Disease

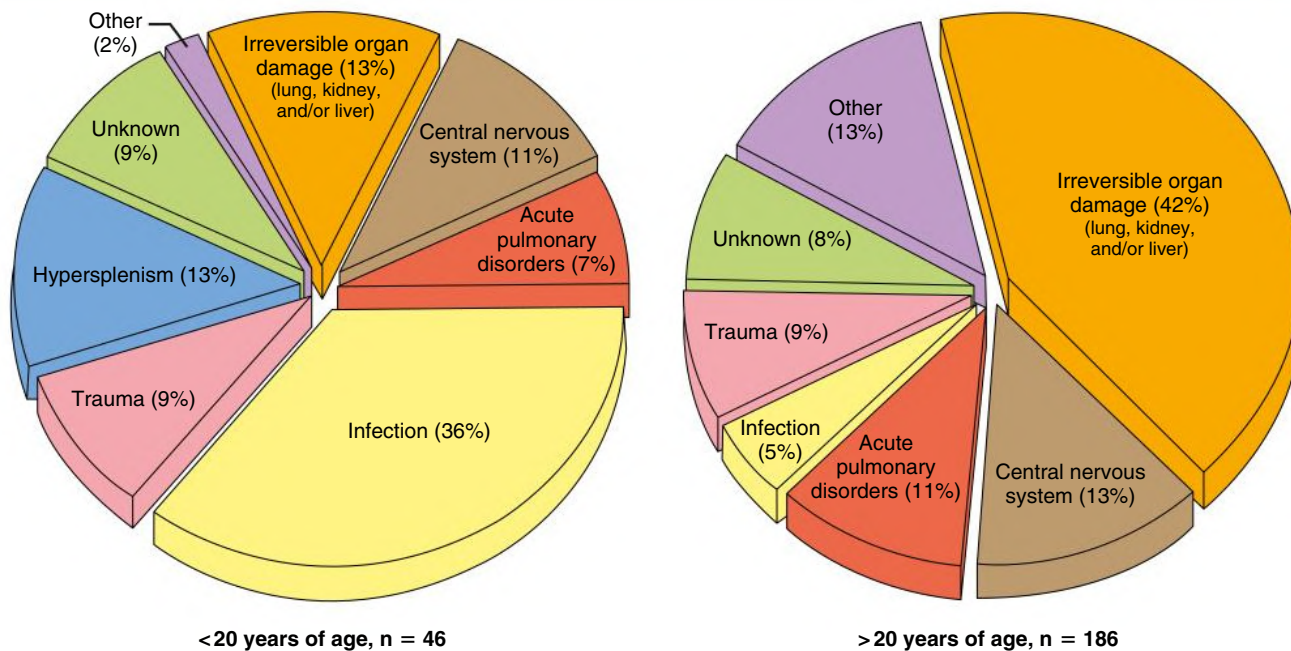


Fig. 51.4 Mortality in Sickle Cell Disease. Causes of death among 232 HbSS patients, comparing patients <20 years (46 died) with patients >20 years (186 died). Infection category includes both bacterial and viral diseases. (Modified from Powars DR, Chan LS, Hiti A, et al. Outcome of sickle cell anemia: a 4-decade observational study of 1056 patients. *Medicine (Baltimore)*. 2005;84:363–376.)

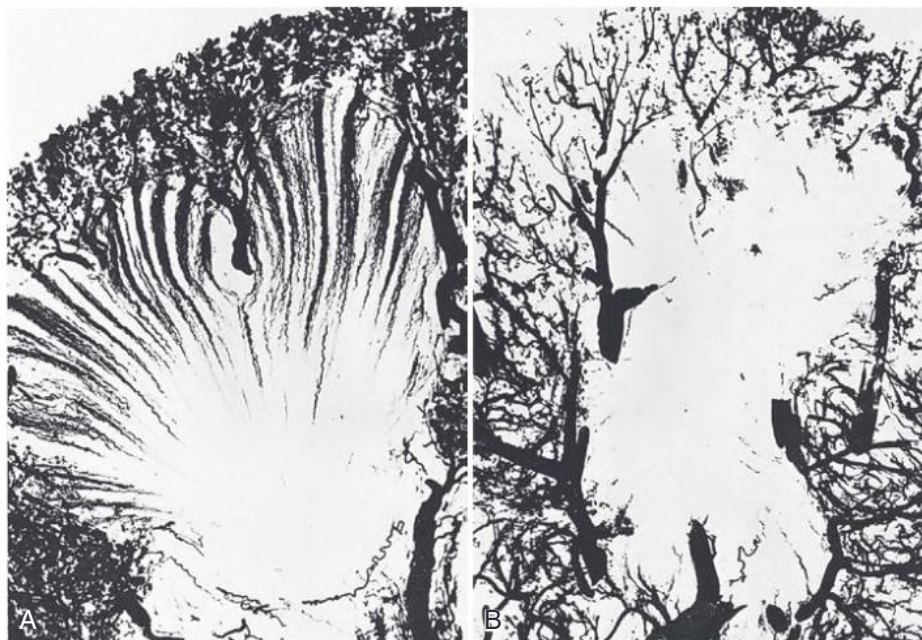


Fig. 51.5 Microradiangiography Showing Loss of Vasa Recta in Sickle Cell Nephropathy. (A) Kidney from control, with normal vasa recta. (B) Patient with sickle cell anemia, with the absence of the vasa recta. (From Statius van Eps LW, Pinedo-Veels C, de Vries GH, et al. Nature of concentrating defect in sickle-cell nephropathy. Microradiangiographic studies. *Lancet*. 1970;1:450–452.)

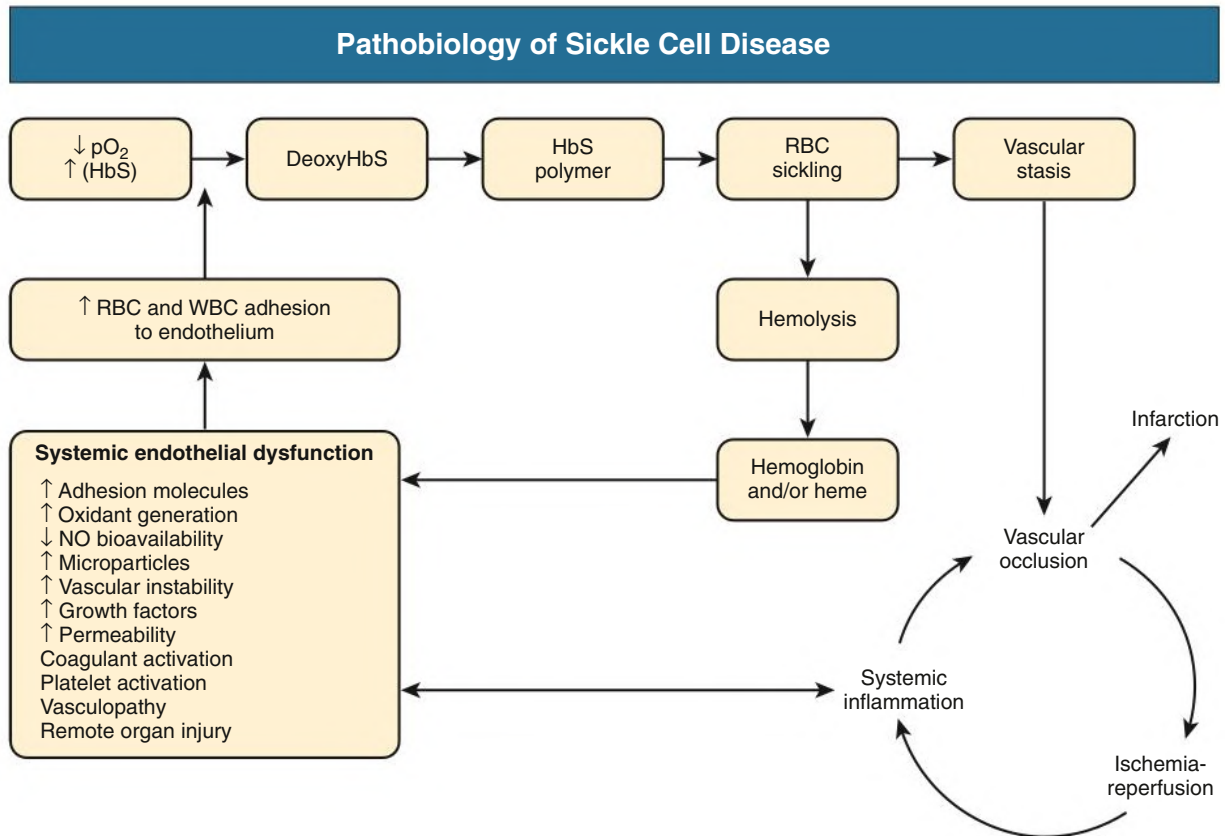


Fig. 51.6 Pathobiology of sickle cell disease. *deoxyHbS*, Deoxygenated HbS; *HbS*, sickle hemoglobin; *NO*, nitric oxide; *pO₂*, partial pressure of oxygen; *RBC*, red blood cell; *WBC*, white blood cell. (Modified from Nath KA, Hebbel RP. Sickle cell disease: renal manifestations and mechanisms. *Nat Rev Nephrol*. 2015;11:161–171.)

The endothelial dysfunction and vasculopathic state of SCD described earlier is also pivotal to the development of SCN. Crosstalk between glomerular endothelial cells and podocytes is important to maintain healthy glomerular function; when the endothelium is dysfunctional, this could lead to podocyte injury and the development of glomerular lesions and proteinuria. FMS-like tyrosine kinase 1 is a receptor for vascular endothelial growth factor (VEGF), and increased levels of the soluble form (sFlt-1) have been associated with endothelial dysfunction in preeclampsia. Serum levels of sFlt-1 are raised in SCD and so may play a role in the development of proteinuria.²²

CLINICAL MANIFESTATIONS OF SICKLE CELL NEPHROPATHY

Glomerular Abnormalities

Hyperfiltration

Glomerular hypertrophy is ubiquitous in SCD and has been reported in children as young as 1 to 3 years of age (Fig. 51.7).²³ Increased kidney growth is observed from infancy in children with SCD and accompanies the early rise in GFR.^{24,25} Glomerular filtration rate continues to rise throughout childhood and early adulthood, often exceeding 200 mL/min/1.73 m². Although this hyperfiltration is not associated with an overt increase in systemic blood pressure (BP), recent studies suggest children with SCD may have a degree of masked hypertension which can be detected with ambulatory monitoring.^{26,27} Persistent hyperfiltration can eventually lead to damage to the glomerular basement membrane and nephrosclerosis, feeding into a cycle of increasing single-nephron GFR, further nephron loss, and an overall decline in GFR during adulthood.

GFR in Children With Sickle Cell Disease

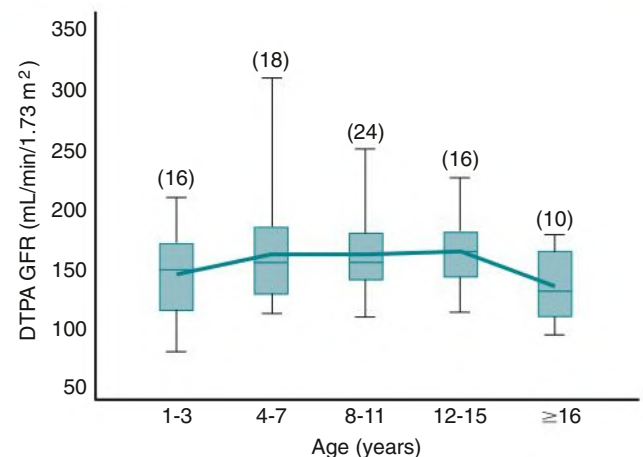


Fig. 51.7 Diethylenetriaminepentaacetic acid (DTPA) glomerular filtration rate (GFR) in children with sickle cell disease. (From Aygun B, Mortier NA, Smeltzer MP, et al. Glomerular hyperfiltration and albuminuria in children with sickle cell anemia. *Pediatr Nephrol*. 2011;26:1285–1290.)

Albuminuria and Proteinuria

Albuminuria occurs after a period of prolonged hyperfiltration and is apparent in some patients from late childhood.²⁸ Unlike hyperfiltration, the prevalence of albuminuria continues to increase with

age, reaching more than 60% of all patients with HbSS SCD over the age of 45 years.^{13,29} In some patients, albuminuria progresses and occasionally reaches the nephrotic range (proteinuria >3.5 g/24 h), which is associated with increased mortality.¹⁵ Nephrotic syndrome, although uncommon (~4% of all those with proteinuria), is associated with a very poor kidney prognosis. One rare cause of sudden onset of nephrotic syndrome in patients with SCD is recent infection with human parvovirus B19 (HPV B19). Although widespread in the community, this usually benign infection becomes significant in patients with SCD because it causes acute but self-limiting red cell aplasia. In combination with severe hemolysis, this leads to life-threatening anemia requiring supportive transfusion. Several reports have described patients in whom infection was followed by acute nephrotic syndrome within 2 to 3 months. In cases in which biopsy has been performed early, the collapsing variant of focal segmental glomerulosclerosis (FSGS) has been found (with or without evidence of direct HPV B19 infection). Although the acute features of the nephrotic syndrome are self-limiting, the sequelae of the condition are worsening glomerulosclerosis, tubulointerstitial fibrosis, and progressive kidney dysfunction.³⁰

Tubular Abnormalities

Hyposthenuria

Hyposthenuria (the excretion of urine of low specific gravity secondary to inability of the kidney to concentrate the urine) is almost universal in people with SCD (who can rarely achieve a urine osmolality above 450 mOsm/kg, even under water-deprived conditions) and also occurs in older people with sickle cell trait. Hyposthenuria often leads to enuresis in children and can cause marked dehydration. It is primarily caused by sickling in the vasa recta, leading to microthrombi, infarction, and collateral formation of blood vessels. As a consequence, there is a defect in the countercurrent exchange mechanism resulting in insufficient trapping of solute in the inner medulla and leading to abnormalities in water conservation.¹⁹ In addition, the increased delivery of salt and water in the tubular filtrate secondary to the high GFR and the intermittent hypoxia caused by sickling in the microvasculature lead to local endothelin-1 release. Endothelin-1 (ET-1) is a potent vasoconstrictor and also has marked natriuretic and diuretic properties through stimulation of ET type b receptors in the kidney collecting ducts, leading to increased salt and water loss.³¹ Although hyposthenuria is reversible by blood transfusion until the age of 10 years, after this age it becomes irreversible and is associated with a permanently damaged microvasculature and increased urine output, leading to a tendency to dehydration.^{19,32}

Increased Proximal Tubular Function

The increase in sodium and water loss from the nephron leads to a reactive increase in sodium and water reabsorption by the proximal tubule secondary to tubuloglomerular feedback. This reabsorption of sodium is the driving force for the reabsorption of other solutes such as phosphate and β_2 -microglobulin, and hence many patients have hyperphosphatemia. There is also an increase in proximal tubular secretion of other solutes, such as creatinine and uric acid. Despite this, serum levels of uric acid are often raised as a result of severe hemolysis and may result in episodes of acute gout. In contrast, up to 30% of the total creatinine excretion can arise from tubular secretion resulting in low serum creatinine levels and overestimation of GFR when creatinine-based formulas are used.³³ Increased sodium reabsorption requires increased oxygen consumption and hypermetabolism of the kidney tubules, a phenomenon that may exacerbate kidney hypoxia in SCD.²²

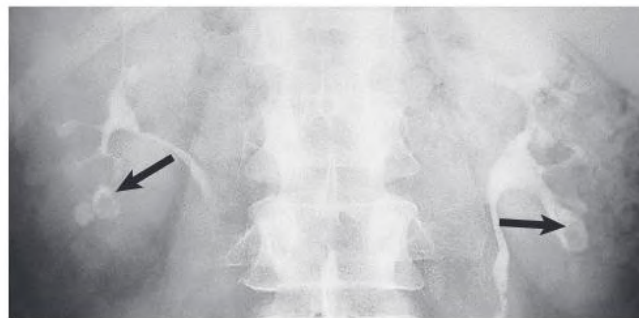


Fig. 51.8 Papillary Necrosis in Sickle Cell Disease. Intravenous urography shows abnormal calyces with filling defects (arrows).

Acidification Defect

A retrospective study in 2014 of 411 homozygous SCD patients with eGFR greater than 60 mL/min/1.73 m² revealed that 42% had partial metabolic acidosis as assessed by serum levels of carbon dioxide (52% of women, 27% of men), apparently because of a lack of ammonia buffering capacity rather than a defect in the distal tubule. Although serum potassium levels were not linked to acidosis in this study, hyperkalemia is a common phenomenon in patients with SCD and has been attributed by others to hyperchloremic metabolic acidosis linked to type IV kidney tubular acidosis or resistance of the distal tubule to aldosterone.²²

Hematuria, Papillary Necrosis, and Kidney Medullary Carcinoma

Hematuria is common in patients with SCD and sickle cell trait. It can range from painless microhematuria, through visible and painless, to visible and painful. It is usually self-limiting but occasionally can be severe enough to require transfusion. Microinfarcts, capillary congestion, and rupture are often the cause, but occasionally, complete occlusion of the vasa recta can lead to papillary necrosis with sloughing of the ischemic papilla, severe hemorrhage, and obstruction, which may be complicated by superimposed infection and painful clot colic (Fig. 51.8).¹⁷ The left-sided predominance of hematuria has been attributed to the so-called nutcracker phenomenon, with compression of the left renal vein between the aorta and the superior mesenteric artery, increasing the pressure in the renal vein. This may contribute to the development of hematuria in patients with SCD, because the increased renal vein pressure could worsen anoxia in the kidney medulla, increasing the likelihood of sickling in the left kidney.

Renal medullary carcinoma (RMC) is an aggressive form of renal cell carcinoma that uniquely affects patients with sickle cell hemoglobinopathies, particularly sickle cell trait or HbSC, especially in teenagers and young adults. It is similar to but distinct from renal cell carcinoma with medullary phenotype (RCC-MP), which is found in individuals who do not carry a sickle gene, but both are characterized by loss of protein expression of the tumor suppressor gene *SMARCB1*. Chronic medullary hypoxia is thought to contribute to its pathogenesis. The tumors are resistant to chemotherapy and tend to be metastatic at diagnosis, with a reported postsurgical mean survival of only 15 weeks. It is not yet clear whether regular evaluation for RMC in young patients with SCD or trait could result in an early diagnosis and a better survival. Gross hematuria, flank pain, and weight loss are ominous signs of malignancy, particularly in young patients with sickle cell trait. The tumor is typically located deep in the parenchyma, unlike Wilms tumor or renal cell carcinoma.³⁴

URINARY TRACT INFECTIONS

Patients with SCD have an increased susceptibility to bacterial infections; even low-grade bacteremia with a common organism may be fatal. In addition to the impaired immunity resulting from auto-splenectomy, there is opsonic antibody deficiency, which predisposes to bacterial infections. Bacteriuria was found to be present in 26% of children with SCD presenting with fever in Nigeria, and urinary tract infection complicated pregnancy in 12% of mothers with SCD in a large UK cohort.^{35,36}

Pyelonephritis and urosepsis, as with any infection, may precipitate a sickle cell crisis. The most common organisms isolated include *Escherichia coli*, *Klebsiella* spp, and other gram-negative Enterobacteriaceae.

CLINICAL SYNDROMES OF KIDNEY IMPAIRMENT

Acute Kidney Injury

Although much is known about CKD, there is less in the literature about AKI in patients with SCD. It is reported as a complicating factor in 2% to 8% of hospital admissions with painful vaso-occlusive crisis or acute chest syndrome. The severity of the AKI appears to be directly related to the severity of the acute sickling crisis. Other causes of AKI are rhabdomyolysis, sepsis, and drug nephrotoxicity. Less common causes of acute kidney dysfunction are kidney vein thrombosis and hepatorenal syndrome (caused by SCD-related hepatic failure). Volume depletion due to inability to concentrate urine makes patients more susceptible to AKI. Although most patients recover, repeated episodes of AKI increase the risk for CKD. After severe episodes, GFR on recovery is often lower than before the acute event.³ In addition, patients with underlying CKD are more prone to AKI, making this a vicious cycle in patients with frequent hospital admissions.^{37,38}

Progressive Chronic Kidney Disease

Progressive loss of kidney function in SCN is an increasingly large problem as life expectancy increases. Similar to diabetic nephropathy, patients with SCN progress through stages of tubular dysfunction and hyperfiltration, through mild to moderate albuminuria to severe albuminuria, and eventually loss of GFR. Over time, single-nephron GFR increases as other nephrons are lost, resulting in worsening damage to the glomeruli (manifested by FSGS), increasing albuminuria, and subsequent interstitial fibrosis and tubular atrophy. Although hypertension is less common in patients with SCD than in an age- and ethnicity-matched cohort, when it is present it has a marked impact on the rate of progression of CKD.³⁹ For this reason, it may be necessary to aim for lower blood pressure targets in these patients than might be advocated in people without SCD.

Predictors of CKD progression are hypertension, proteinuria, nephrotic syndrome, hematuria, increasingly severe anemia, and inheritance of the Bantu, or Central African Republic, β -globin gene cluster haplotype. In addition, younger age at diagnosis and higher duration of SCD were found to strongly predict the development of nephropathy.⁴⁰

As GFR declines, the ability of the kidney to synthesize erythropoietin (EPO) also declines. Chronic anemia and tissue hypoxia are strong drivers for EPO synthesis, and patients with SCD with normal kidney function often have EPO levels well above the normal range. However, when the GFR falls below approximately 60 mL/min/1.73 m², their ability to produce sufficient endogenous EPO also begins to decline, resulting in worsening anemia. In patients who develop progressive kidney dysfunction secondary to SCN, the rate of decline can be quite

rapid once the GFR falls below 40 mL/min/1.73 m², and so timely preparation for kidney replacement therapy (KRT) is very important. Many patients who have suffered frequent admissions to hospital often have very poor peripheral veins and so need expert surgical input when planning for dialysis access.

INVESTIGATION AND MANAGEMENT OF SICKLE CELL NEPHROPATHY

Serum cystatin C is promising as a more accurate marker of kidney function than creatinine in adults and children with SCD and may detect decline in GFR earlier.²⁶ However, it has yet to be fully validated against gold standard measures that calculate true GFR in SCD patients, so creatinine and creatinine-based equations continue to be used in the clinic. It is therefore the pattern and rate of change of either serum creatinine or eGFR that should be considered rather than the absolute value, bearing in mind that relatively modest changes in creatinine at higher levels of GFR can represent significant decline in kidney function. Kidney function and albuminuria should be monitored at least annually.⁴¹ Creatinine levels are often low in people with SCD secondary to reduced muscle bulk, hyperfiltration, and increased proximal tubular secretion, resulting in a high eGFR. Increased rate of change of creatinine may therefore indicate declining GFR before the value moves out of the normal range. Anyone with an eGFR that is declining at more than 5 mL/min/yr or an absolute value less than 60 mL/min should be identified and referred to a nephrologist (Fig. 51.9). Rigorous BP control is recommended, as for other causes of CKD.³⁹ A BP of 140/90 is currently permissible for patients with a negative dipstick result or those with a urine albumin-to-creatinine ratio (uACR) less than 3.5 mg/mmol, but a target of 130/80 should be used for those with a uACR greater than 3.5 mg/mmol.⁴² Urinary tract infections should be promptly treated with appropriate antibiotics. Glucose-6-phosphate dehydrogenase deficiency (G6PD deficiency), which may precipitate intravascular hemolysis, should be excluded because some antibiotics, including sulfonamides and nitrofurantoin, precipitate hemolysis by antagonizing folate synthesis. Long-term use of non-steroidal antiinflammatory drugs should be avoided in patients with eGFR less than 60 mL/min/1.73 m²; if unavoidable, regular monitoring of kidney function is recommended.

The population at risk for developing SCN is also at risk for other diseases that affect the kidneys, including lupus nephritis, various forms of glomerulonephritis, blood-borne viruses, renal carcinoma, myeloma, and kidney stones. This differential should be considered when investigating a patient with SCD and new-onset proteinuria or hematuria. Although it is common practice not to investigate patients with albuminuria (uACR >3.5 mg/mmol), patients with urinary protein-to-creatinine ratio (uPCR) greater than 50 mg/mmol should be evaluated for other causes of CKD (Fig. 51.9); if any of these investigations are abnormal, or the patient's signs and symptoms do not conform to those expected within the natural history of SCN as described previously, they should be referred for further evaluation. In particular, patients with sudden onset of heavy proteinuria, with or without nephrotic syndrome, warrant kidney biopsy to look for causes other than SCN. There is no pathognomonic lesion that defines SCN. Glomerular hypertrophy with distended capillaries is universally found but is not confined to those who have developed albuminuria or proteinuria. FSGS is the most common lesion associated with proteinuria, but it is not specific to SCN (Fig. 51.10). Other lesions that have been reported on kidney biopsy in SCD include thrombotic microangiopathy and membranoproliferative glomerulonephritis; neither is exclusive to SCN. The only characteristic interstitial lesion is abundant

Investigation and Management of Hematuria and Proteinuria in Sickle Cell Disease

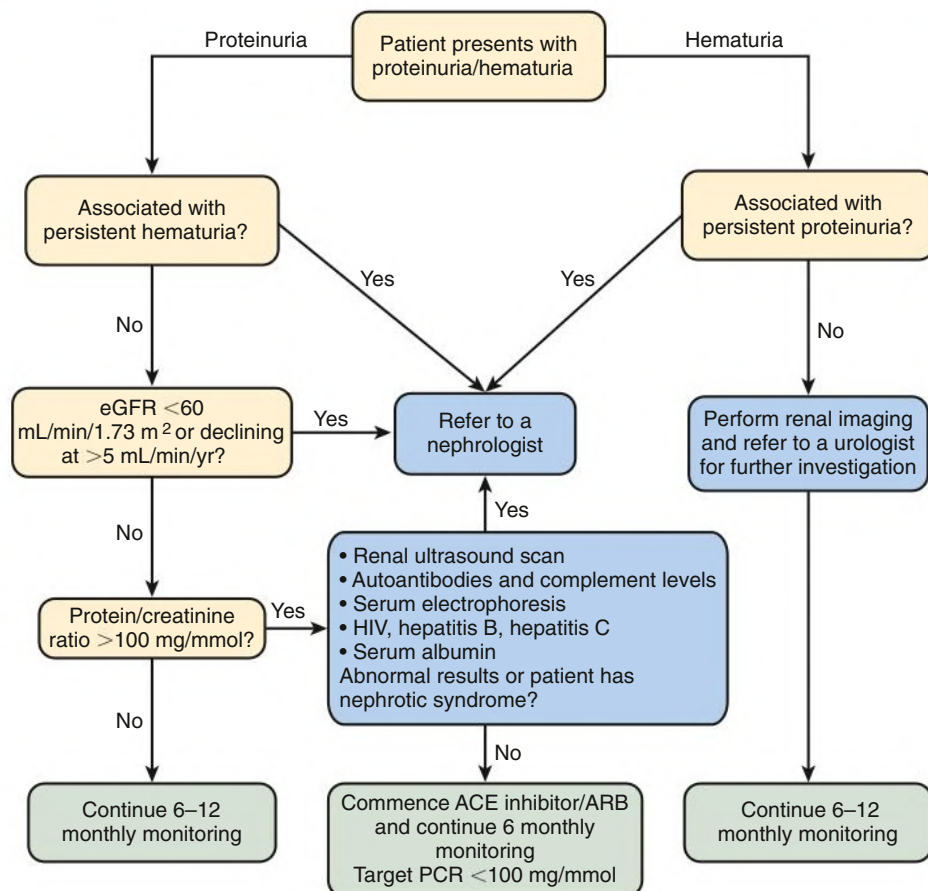


Fig. 51.9 Investigation and management of hematuria/proteinuria in patients with sickle cell disease. ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; eGFR, estimated glomerular filtration rate; PCR, protein-to-creatinine ratio.

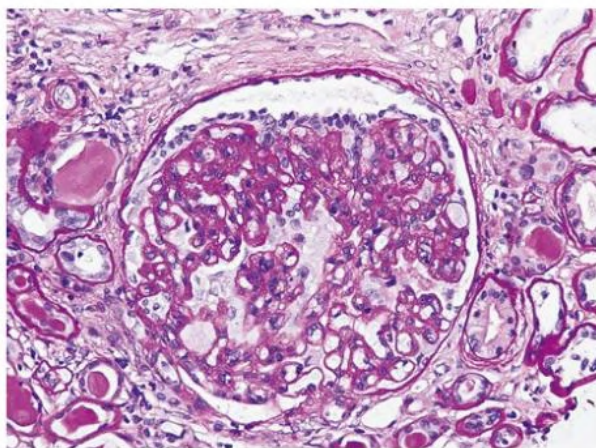


Fig. 51.10 Focal segmental glomerulosclerosis in sickle cell nephropathy. There is segmental sclerosis in the upper half of the glomerulus. (Courtesy Professor J. Weening.)

hemosiderin granules in proximal tubular epithelial cells.⁴³ Kidney iron deposition also has been noted on magnetic resonance scans in patients with SCD but appears not to be related to liver iron concentration, a marker of total body iron load. Kidney iron does appear to be correlated with markers of hemolysis but has not been shown to be associated with kidney dysfunction or degree of albuminuria.⁴⁴

Inhibition of the Renin-Angiotensin System

Only one small randomized, controlled trial evaluated angiotensin-converting enzyme (ACE) inhibition in patients with SCD and proteinuria; a more recent systematic review concluded that available evidence was insufficient to offer recommendations on treatment.⁴⁵ However, based on the evidence available for other causes of proteinuric kidney disease, ACE inhibitors or angiotensin receptor blockers (ARBs) are recommended for the treatment of adults and children with SCD and albuminuria.⁴¹ These drugs must be introduced cautiously because many patients have a low or normal BP, and so moderate doses can cause postural hypotension, which can be partially circumvented by

taking the medication at bedtime. Patients have reported that this has the benefit of reducing nocturnal urinary frequency, presumably as a direct result of a functional drop in GFR. Patients with SCN are prone to hyperkalemia, which can be exacerbated by ACE inhibitor and ARB treatment and is often the dose-limiting factor. It is also important to inform patients and other caregivers that ACE inhibitors and ARB should be temporarily stopped during acute illnesses associated with dehydration to mitigate the risk for AKI.

Sickle Cell Disease Therapies and Blood Transfusion

Oral hydroxycarbamide (HC; also known as hydroxyurea) was the first licensed drug for the management of SCD. Although HC has pleiotropic effects, it primarily acts to increase levels of fetal Hb that serve to dilute the levels of HbS and reduce risk for polymerization. However, there are no RCTs of HC use in the management or progression of SCN in adults and only one in infants, which showed no effect on hyperfiltration, though urinary concentrating ability was improved.⁴⁶ A number of observational trials of HC in older children and adults have shown a reduction in hyperfiltration and albuminuria.³ HC therefore should be considered for patients with microalbuminuria/proteinuria or abnormal GFR together with an ACE inhibitor or ARB (or in those intolerant of this treatment). Over the past 5 years there have been a number of new treatments for SCD, some of which are now licensed for use in preventing VOC. Although none has yet to be proven to affect the progression of CKD, trials in this area are planned.⁴⁷

There is little evidence for the benefits of long-term RBC transfusions on prevention of kidney complications of SCD. A retrospective analysis of 120 children with sickle hemoglobinopathies concluded that chronic transfusion protected against the onset of albuminuria when started before the age of 9 years. Two studies, however, found that chronic transfusion made no difference to onset of albuminuria and that prolonged courses of transfusion therapy can lead to iron overload, which is particularly difficult to treat in patients with CKD.³

Erythropoiesis-Stimulating Agents

Erythropoiesis-stimulating agents (ESAs) can be useful, particularly in combination with hydroxycarbamide in patients who are intolerant of hydroxycarbamide alone because of reticulocytopenia. They should be commenced when the Hb has fallen by approximately 10% to 15% from the normal baseline at steady state, though patients with CKD stages 3 to 4 often require very high doses of ESAs to have an impact on Hb levels. Although Hb targets should be lower than in the general population with CKD (80–100 g/dL) because of the increased risk for triggering VOC, they are rarely achieved, and most patients become transfusion dependent by the time they reach ESKD. However, it is often still beneficial to continue ESA therapy after the commencement of KRT because this can prolong the interval between RBC transfusion and minimize the risks for iron overload.³ It is important to ensure that adequate iron stores are maintained to achieve maximum erythropoiesis. Intestinal losses of iron resulting from subclinical bleeding are significant in advanced CKD, and absorption of oral iron is reduced. Intravenous iron supplementation may be necessary therefore in patients deficient in iron, on ESAs, and not receiving iron through regular transfusions.

Hemopoietic Stem Cell Transplantation

Hemopoietic stem cell transplantation (HSCT) is potentially curative and may reduce hyperfiltration in the young⁴⁸ but has largely been limited to children with severe cerebrovascular complications, acute

chest syndrome, or frequent VOC not responding to hydroxycarbamide therapy. The few reports of HSCT in adults with SCN have mixed kidney outcomes.³

KIDNEY REPLACEMENT THERAPY

Dialysis

Outcome data for patients with SCD on dialysis are few, but an early report from the United States showed that the average age of those reaching ESKD was very young (23.1 years in patients with HbSS), and the mean time to death after was only 4 years despite regular hemodialysis.⁴⁹ More recent data from France and the United States report that the 5-year mortality rate for patients with SCD with ESKD remains significantly higher than for patients without SCD. They have more complications, are admitted to hospital more frequently and are less likely to be offered a kidney transplant.^{50,51} Infections and thrombosis of dialysis access were commonly encountered complications in the group with SCD.

Transplantation

Although there may be many obstacles in the path to kidney transplantation, it is probably the preferred modality for patients with SCD requiring KRT. Although long-term graft and patient survival are not quite as good as for patients with other causes of ESKD, the prognosis for individuals with SCN is far better after transplantation with a projected 7-year survival of 67% (vs. 83% for other African Americans) compared with a 10-year survival of only 14% for those who remain on dialysis.²¹ Although outcomes after transplantation in SCD have improved over the last 20 years, it is not without complications. Delayed graft function and graft loss are more common than in other patient groups, and the recurrence of frequent VOC after transplantation as the Hb rises is problematic.⁵² Treatment with regular exchange transfusion aimed at keeping the HbS level below 30% may help reduce the risk for these complications and does not appear to be associated with increased rejection rates but is associated with improved patient survival.⁵³

SICKLE CELL TRAIT AND CHRONIC KIDNEY DISEASE

In patients with heterozygous sickle cell trait, approximately 40% of RBC Hb is HbS, and the rest is normal HbA. In general, these patients have normal Hb levels and do not have symptoms of hemolysis or vaso-occlusion. However, there have been reports of catastrophic VOC and sudden death in young people with sickle cell trait under extreme adverse conditions such as excessive exercise or exposure to severe hypoxia. In addition, renal medullary carcinoma is more common in patients with SCT than SCD, though the reasons for this are unclear. Nonmalignant microhematuria and macrohematuria are reported more frequently in patients with sickle cell trait than in the general population, and older patients exhibit a loss of urinary concentrating ability. Patients who coinherit sickle cell trait and adult polycystic kidney disease (APKD) have a more rapid decline to ESKD than family members with APKD who do not carry an HbS gene. Whether having sickle cell trait alone is a risk factor for progressive CKD has been debated, but a recent population-based study in the United States has shown that SCT is associated with both CKD and progression to ESKD (hazard ratio 2.03), independent of APOL1 status.⁵⁴

SELF-ASSESSMENT QUESTIONS

1. Patients with SCD:
 - A. often develop albuminuria before they are 5 years old.
 - B. have reduced urinary concentrating capacity.
 - C. have reduced total kidney blood flow in childhood.
 - D. are often hypokalemic.
2. Patients with SCD and ESKD:
 - A. have a lower mortality rate when on dialysis than patients with adult polycystic kidney disease.
 - B. who have a kidney transplant should avoid blood transfusion, as it is associated with increased rejection rates.
 - C. who are on dialysis should be prescribed erythropoietin, as this usually avoids the need for blood transfusion.
 - D. have a higher rate of delayed graft function after kidney transplantation than patients with ESKD of other causes.
3. Hydroxycarbamide:
 - A. should be avoided in patients with chronic kidney disease.
 - B. has been shown to reduce hyperfiltration in children with SCD in a randomized controlled trial.
 - C. works by increasing fetal hemoglobin production.
 - D. should not be given in combination with inhibitors of the renin-angiotensin system.

REFERENCES

- Piel FB, Patil AP, Howes RE, et al. Global epidemiology of sickle haemoglobin in neonates: a contemporary geostatistical model-based map and population estimates. *Lancet*. 2013;381:142–151.
- Piel FB, Tatem AJ, Huang Z, et al. Global migration and the changing distribution of sickle haemoglobin: a quantitative study of temporal trends between 1960 and 2000. *Lancet Glob Health*. 2014;2:e80–e89.
- Sharpe CC, Thein SL. How I treat renal complications in sickle cell disease. *Blood*. 2014;123:3720–3726.
- Hockham C, Piel FB, Gupta S, et al. Understanding the contrasting spatial haplotype patterns of malaria-protective beta-globin polymorphisms. *Infect Genet Evol*. 2015;36:174–183.
- Serjeant GR. The natural history of sickle cell disease. *Cold Spring Harb Perspect Med*. 2013;3:a011783.
- Akinsheye I, Solovieff N, Ngo D, et al. Fetal hemoglobin in sickle cell anemia: molecular characterization of the unusually high fetal hemoglobin phenotype in African Americans. *Am J Hematol*. 2012;87:217–219.
- Rees DC, Williams TN, Gladwin MT. Sickle-cell disease. *Lancet*. 2010;376:2018–2031.
- Potoka KP, Gladwin MT. Vasculopathy and pulmonary hypertension in sickle cell disease. *Am J Physiol Lung Cell Mol Physiol*. 2015;308:L314–L324.
- Platt OS, Brambilla DJ, Rosse WF, et al. Mortality in sickle cell disease. Life expectancy and risk factors for early death. *N Engl J Med*. 1994;330:1639–1644.
- Davies SC, Oni L. Management of patients with sickle cell disease. *BMJ*. 1997;315:656–660.
- Mack AK, Thompson AA. Primary and secondary stroke prevention in children with sickle cell disease. *J Pediatr Health Care*. 2016.
- Powars DR, Chan LS, Hiti A, et al. Outcome of sickle cell anemia: a 4-decade observational study of 1056 patients. *Medicine (Baltim)*. 2005;84:363–376.
- Guasch A, Navarrete J, Nass K, et al. Glomerular involvement in adults with sickle cell hemoglobinopathies: prevalence and clinical correlates of progressive renal failure. *J Am Soc Nephrol*. 2006;17:2228–2235.
- Serjeant GR, Serjeant BE, Mason KP, et al. The changing face of homozygous sickle cell disease: 102 patients over 60 years. *Int J Lab Hematol*. 2009;31:585–596.
- Drawz P, Ayyappan S, Nouria M, et al. Kidney disease among patients with sickle cell disease, hemoglobin SS and SC. *Clin J Am Soc Nephrol*. 2016;11:207–215.
- Saraf SL, Shah BN, Zhang X, et al. APOL1, alpha-thalassemia, and BCL11A variants as a genetic risk profile for progression of chronic kidney disease in sickle cell anemia. *Haematologica*. 2017;102:e1–e6.
- Sharpe CC, Thein SL. Sickle cell nephropathy - a practical approach. *Br J Haematol*. 2011;155:287–297.
- Evans RG, Gardiner BS, Smith DW, et al. Intrarenal oxygenation: unique challenges and the biophysical basis of homeostasis. *Am J Physiol Renal Physiol*. 2008;295:F1259–F1270.
- Stadius van Eps LW, Pinedo-Veels C, de Vries GH, et al. Nature of concentrating defect in sickle-cell nephropathy. Microradioangiographic studies. *Lancet*. 1970;1:450–452.
- Gurbanov K, Rubinstein I, Hoffman A, et al. Differential regulation of renal regional blood flow by endothelin-1. *Am J Physiol*. 1996;271:F1166–F1172.
- Scheinman JI. Sickle cell disease and the kidney. *Nat Clin Pract Nephrol*. 2009;5:78–88.
- Nath KA, Hebbel RP. Sickle cell disease: renal manifestations and mechanisms. *Nat Rev Nephrol*. 2015;11:161–171.
- Aygun B, Mortier NA, Smeltzer MP, et al. Glomerular hyperfiltration and albuminuria in children with sickle cell anemia. *Pediatr Nephrol*. 2011;26:1285–1290.
- McCarville MB, Luo Z, Huang X, et al. Abdominal ultrasound with scintigraphic and clinical correlates in infants with sickle cell anemia: baseline data from the BABY HUG trial. *AJR Am J Roentgenol*. 2011;196:1399–1404.
- Ware RE, Rees RC, Sarnaik SA, et al. Renal function in infants with sickle cell anemia: baseline data from the BABY HUG trial. *J Pediatr*. 2010;156:66–70.e61.
- Lebensburger JD, Cutter GR, Howard TH, et al. Evaluating risk factors for chronic kidney disease in pediatric patients with sickle cell anemia. *Pediatr Nephrol*. 2017;32:1565–1573.
- Moodalail DG, Falkner B, Keith SW, et al. Ambulatory hypertension in a pediatric cohort of sickle cell disease. *J Am Soc Hypertens*. 2018;12:542–550.
- Lebensburger JD, Aban I, Pernell B, et al. Hyperfiltration during early childhood precedes albuminuria in pediatric sickle cell nephropathy. *Am J Hematol*. 2019;94:417–423.
- Day TG, Drasar ER, Fulford T, et al. Association between hemolysis and albuminuria in adults with sickle cell anemia. *Haematologica*. 2012;97:201–205.
- Quek L, Sharpe C, Dutt N, et al. Acute human parvovirus B19 infection and nephrotic syndrome in patients with sickle cell disease. *Br J Haematol*. 2010;149:289–291.
- Nakano D, Pollock D. New concepts in endothelin control of sodium balance. *Clin Exp Pharmacol Physiol*. 2011.
- Stadius van Eps LW, Schouten H, La Porte-Wijsman LW, et al. The influence of red blood cell transfusions on the hyposthenuria and renal hemodynamics of sickle cell anemia. *Clin Chim Acta*. 1967;17:449–461.
- Thompson J, Reid M, Hambleton I, et al. Albuminuria and renal function in homozygous sickle cell disease: observations from a cohort study. *Arch Intern Med*. 2007;167:701–708.
- Msaouel P, Hong AL, Mullen EA, et al. Updated recommendations on the diagnosis, management, and clinical trial eligibility criteria for patients with renal medullary carcinoma. *Clin Genitourin Cancer*. 2019;17:1–6.
- Mava Y, Ambe JP, Bello M, et al. Urinary tract infection in febrile children with sickle cell anaemia. *West Afr J Med*. 2011;30:268–272.
- Oteng-Ntim E, Ayensah B, Knight M, et al. Pregnancy outcome in patients with sickle cell disease in the UK—a national cohort study comparing sickle cell anaemia (HbSS) with HbSC disease. *Br J Haematol*. 2015;169:129–137.
- Lebensburger JD, Palabindela P, Howard TH, et al. Prevalence of acute kidney injury during pediatric admissions for acute chest syndrome. *Pediatr Nephrol*. 2016;31:1363–1368.
- McCormick M, Richardson T, Warady BA, et al. Acute kidney injury in paediatric patients with sickle cell disease is associated with increased morbidity and resource utilization. *Br J Haematol*. 2020;189:559–565.
- Gordeuk VR, Sachdev V, Taylor JG, et al. Relative systemic hypertension in patients with sickle cell disease is associated with risk of pulmonary hypertension and renal insufficiency. *Am J Hematol*. 2008;83:15–18.
- Arogundade FA, Sanusi AA, Hassan MO, et al. An appraisal of kidney dysfunction and its risk factors in patients with sickle cell disease. *Nephron Clin Pract*. 2010;118:c225–c231.
- Liem RI, Lanzkron S, DC T, et al. American Society of Hematology 2019 guidelines for sickle cell disease: cardiopulmonary and kidney disease. *Blood Adv*. 2019;3:3867–3897.
- Stevens PE, Levin A. Kidney disease: improving Global Outcomes Chronic Kidney Disease Guideline Development Work Group M. Evaluation and management of chronic kidney disease: synopsis of the kidney disease: improving global outcomes 2012 clinical practice guideline. *Ann Intern Med*. 2013;158:825–830.
- Zahr RS, Yee ME, Weaver J, et al. Kidney biopsy findings in children with sickle cell disease: a Midwest Pediatric Nephrology Consortium study. *Pediatr Nephrol*. 2019;34:1435–1445.
- Vasavda N, Gutierrez L, House MJ, et al. Renal iron load in sickle cell disease is influenced by severity of haemolysis. *Br J Haematol*. 2012;157:599–605.
- Sasongko TH, Nagalla S, Ballas SK. Angiotensin-converting enzyme (ACE) inhibitors for proteinuria and microalbuminuria in people with sickle cell disease. *Cochrane Database Syst Rev*. 2013;CD009191.
- Wang W, Thompson B, investigators BH. Hydroxyurea treatment of infants with sickle cell anaemia: results of the BABY HUG study. *Paediatric Blood and Cancer*. 2010;54:787.

47. Osunkwo I, Manwani D, Kanter J. Current and novel therapies for the prevention of vaso-occlusive crisis in sickle cell disease. *Ther Adv Hematol*. 2020;11:2040620720955000.
48. Pedersen SJV, Monagel DA, Mammen C, et al. Stable renal function in children and adolescents with sickle cell disease after nonmyeloablative hematopoietic stem cell transplantation. *Pediatr Blood Cancer*. 2020;67:e28568.
49. Powars DR, Elliott-Mills DD, Chan L, et al. Chronic renal failure in sickle cell disease: risk factors, clinical course, and mortality. *Ann Intern Med*. 1991;115:614–620.
50. Olaniran KO, Eneanya ND, Zhao SH, et al. Mortality and hospitalizations among sickle cell disease patients with end-stage kidney disease Initiating dialysis. *Am J Nephrol*. 2021:1–9.
51. Nielsen L, Canoui-Poittrine F, Jais JP, et al. Morbidity and mortality of sickle cell disease patients starting intermittent haemodialysis: a comparative cohort study with non-Sickle dialysis patients. *Br J Haematol*. 2016;174:148–152.
52. Gérardin C, Moktefi A, Couchoud C, et al. Survival and specific outcome of sickle cell disease patients after renal transplantation. *Br J Haematol*. 2019;187:676–680.
53. Willis JC, Awogbade M, Howard J, et al. Outcomes following kidney transplantation in patients with sickle cell disease: the impact of automated exchange blood transfusion. *PLoS One*. 2020;15:e0236998.
54. Naik RP, Irvin MR, Judd S, et al. Sickle cell trait and the risk of ESRD in blacks. *J Am Soc Nephrol*. 2017;28:2180–2187.